

ARTIFICIAL INTELLIGENCE

PROJECT REPORT

CONTENT CREATION TOOL

AI-Powered Video Processing & Face Detection

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1. Executive Summary

Content Creation Tool is a comprehensive desktop application engineered to automate and streamline the complete video content creation workflow. Built on Python with a Tkinter-based graphical interface, the platform integrates industry-standard tools — yt-dlp, FFmpeg, MediaPipe, and face_recognition — into a single, unified environment designed for content creators, social media managers, and video production teams.

The application delivers an end-to-end pipeline encompassing video acquisition from any major social platform, intelligent clip segmentation, AI-powered face filtering, real-time batch processing, and an integrated stock media marketplace. A robust licensing and authentication system powered by Firebase ensures secure multi-user access with tiered plan management.

6

Workflow Tabs

3

Stock Media APIs

3

License Tiers

AI

Powered Ranking

2. Project Overview

2.1 Project Identity

Project Name	Content Creation Tool
Version	1.0.0 — Phase 1 Complete
Platform	Windows / macOS / Linux (Desktop)
Language	Python 3.11
GUI Framework	Tkinter + ttk
Database	Firebase Realtime Database
Authentication	Firebase Auth
Report Date	June 2026

2.2 Problem Statement

Content creators and video production teams face significant inefficiencies when managing multi-step video workflows. The process of downloading, segmenting, filtering, and sourcing stock media requires navigating multiple disconnected tools, resulting in substantial time loss, inconsistent output quality, and high operational overhead.

Content Creation Tool addresses these challenges by consolidating the entire workflow into a single, intelligent desktop application — reducing a multi-tool, multi-hour process to a streamlined, automated pipeline.

2.3 Objectives

- Automate video download from all major social media platforms
- Provide intelligent clip segmentation with configurable duration
- Implement AI-powered face detection and recognition filtering
- Deliver a real-time batch processing dashboard with per-clip status
- Integrate a multi-source stock media search engine with AI ranking
- Enforce a secure, Firebase-backed licensing and authentication system
- Maintain full processing history with analytics and export capabilities

3. System Architecture

3.1 Application Structure

Content Creation Tool follows a modular Model-View architecture, separating business logic from the presentation layer. The application is organized into four primary layers:

Layer	Directory	Responsibility
Presentation Layer	gui/	All Tkinter windows, tabs, and widgets
Business Logic	core/	Video operations, filters, utilities
Data & Services	modules/	License management, Firebase, history, stock search
Entry Point	app.py	Application bootstrap, authentication flow

3.2 Directory Structure

```
your_project/ └─ app.py # Main entry point & dashboard └─ core/ |
└─ video_ops.py # yt-dlp download + FFmpeg split | └─ filters.py
# MediaPipe + face_recognition | └─ utils.py # Logging, directory
helpers └─ modules/ | └─ license_manager.py # License validation & caching |
└─ firebase_auth.py # Firebase authentication | └─ firebase_db.py #
Realtime database operations | └─ history_manager.py # Job history (JSON store) |
└─ stock_search.py # Pexels + Pixabay + Unsplash + AI └─ gui/ | └─
license_window.py # License activation screen | └─ video_tool.py # Main
6-tab processing tool | └─ stock_search_window.py # Stock media search UI | └─
preview_player.py # Live clip preview | └─ batch_dashboard.py # Real-time
batch status | └─ admin_panel.py # User management (Admin only) └─
app_history.json # Persistent job history
```

3.3 Authentication & Authorization Flow

1. User launches application → License Activation Window
2. License key validated against Firebase Realtime Database
3. On success → Firebase Login (email + password)
4. Role fetched from Firebase: 'admin' or 'user'
5. Dashboard renders with role-appropriate features
6. Admin: full access + User Management panel
7. User: features determined by license plan tier

4. Features & Modules

4.1 Video Download Module

Powered by yt-dlp, the download module supports over 1,000 websites including YouTube, Instagram, Facebook, TikTok, and Twitter. A custom DownloadController enables pause, resume, and cancel operations. A real-time progress dashboard displays:

- Download percentage (0–100%) with animated progress bar
- Downloaded size vs. total file size in MB
- Live transfer speed (KB/s or MB/s)
- Estimated time remaining (ETA)

4.2 Video Split Module

FFmpeg-based segmentation splits source videos into equal-duration clips. Users configure clip duration via an interactive slider (1–60 seconds). The module supports all major export formats gated by license tier:

Format	Trial	Pro	Lifetime
MP4	✔Yes	✔Yes	✔Yes
AVI	✗No	✔Yes	✔Yes
MKV	✗No	✔Yes	✔Yes
WebM	✗No	✔Yes	✔Yes
MOV	✗No	✔Yes	✔Yes

4.3 Face Filter Module

Two filtering engines are available based on use-case requirements:

- Any Face Detection (MediaPipe): High-speed detection of any human face in clips. Optimal for rapid content triage.
- Specific Face Recognition (face_recognition): Matches clips against a reference image using deep learning embeddings. High accuracy for talent-specific filtering.

A configurable sample rate (every N seconds) balances speed and accuracy. The Batch Dashboard provides real-time per-clip status with color-coded indicators: Processing, Found, Not Found, and Failed.

4.4 Stock Media Search Module

An integrated stock media marketplace aggregates results from three premium free-tier APIs simultaneously:

API	Media Types	Results per Search	Free Limit
Pexels	Photos + Videos	Up to 12 per source	200 req/hour
Pixabay	Photos + Videos	Up to 12 per source	100 req/min
Unsplash	Photos only	Up to 12 per source	50 req/hour

Results are ranked using Google Gemini 2.0 Flash AI based on keyword relevance, visual quality (resolution), media type preference, and tag/title matching. Original-quality downloads are supported with a real-time progress popup displaying speed, MB downloaded, and completion percentage.

4.5 Preview Player

A live preview player allows users to review split clips before processing. Face-matched clips are badged with visual indicators. Navigation supports quick-jump between all clips in the batch.

4.6 Processing History

All processing jobs are automatically logged to a local JSON store (app_history.json). The History tab displays a sortable table with columns for job ID, timestamp, video name, filter type, clip counts, match rate, duration, and export format. Aggregate statistics (total jobs, clips processed, overall match rate) are displayed in summary cards.

5. Licensing & Access Control

5.1 License Tiers

Feature	Trial	Pro	Lifetime
Max Clips per Job	10	500	Unlimited
Export Formats	MP4 only	All 5	All 5
Face Filter	✗	✓	✓
Batch Dashboard	✗	✓	✓
Preview Player	✗	✓	✓
Processing History	✗	✓	✓
Max Devices	1	1	3
Expiry	Session	Date-based	Never

5.2 License Validation Flow

- Key entered by user in activation window
- Key formatted and validated against Firebase Realtime DB
- Device fingerprint checked against allowed device count
- Expiry date verified (Lifetime keys exempt)
- Features dict returned and applied throughout the session
- Successful keys cached locally for auto-login on next launch

5.3 Admin Privileges

Users with the 'admin' role in Firebase receive unrestricted access regardless of license key status. Admin accounts have access to:

- Full Lifetime feature set (enforced independently of license)
- User Management Panel (create, view, and revoke user accounts)
- License Manager (generate keys, set plans, revoke licenses)
- All export formats and unlimited clip processing

6. Technology Stack

Category	Technology	Purpose	Version
Language	Python	Core application runtime	3.11
GUI Framework	Tkinter / ttk	Desktop UI rendering	Built-in
Video Download	yt-dlp	Multi-platform video acquisition	Latest
Video Processing	FFmpeg	Clip segmentation & encoding	6.x
Face Detection	MediaPipe	Any-face detection (fast)	0.10.x
Face Recognition	face_recognition	Specific person matching	1.3.x
Image Processing	Pillow (PIL)	Thumbnail rendering in GUI	11.x
Authentication	Firebase Auth	User login & session management	REST API
Database	Firebase RTDB	License & role storage	REST API
AI Ranking	Gemini 2.0 Flash	Stock media relevance scoring	v1beta
Stock Media	Pexels API	Free stock photos & videos	v1
Stock Media	Pixabay API	Free stock photos & videos	v2
Stock Media	Unsplash API	Free stock photos	v1
HTTP Client	requests	All external API communication	2.x

7. Installation & Setup

7.1 System Requirements

Component	Minimum	Recommended
OS	Windows 10 / macOS 10.15 / Ubuntu 20.04	Windows 11 / macOS 13+
Python	3.9	3.11
RAM	4 GB	8 GB+
Storage	2 GB	10 GB+ (for video downloads)
Internet	Required for download & search features	Broadband (10 Mbps+)

7.2 Python Dependencies

```
pip install yt-dlp pip install opencv-python pip install mediapipe pip install face-recognition pip install Pillow pip install requests pip install firebase-admin
```

7.3 External Dependencies

- FFmpeg: Must be installed and added to system PATH
- CMake + dlib: Required for face_recognition library (Windows)
- Visual C++ Build Tools: Required on Windows for dlib compilation

7.4 Configuration

The following API keys must be configured in modules/stock_search.py before first use:

Key	Source	Free Tier
PEXELS_API_KEY	pexels.com/api	200 requests/hour
PIXABAY_API_KEY	pixabay.com/api/docs	100 requests/minute
UNSPLASH_API_KEY	unsplash.com/developers	50 requests/hour
GEMINI_API_KEY	aistudio.google.com/app/apikey	1,500 requests/day

8. Security Considerations

8.1 Authentication Security

- Firebase Authentication enforces secure email/password login with server-side validation
- ID tokens are stored in memory only — never written to disk in plaintext
- All Firebase Realtime Database reads for user roles require authenticated tokens
- Role-based access control (RBAC) enforced at both UI and feature levels

8.2 License Security

- License keys are validated server-side via Firebase — no local bypass possible
- Device fingerprinting limits concurrent activations per license tier
- License cache stores minimal data; revoked keys are invalidated on next validation
- Admin role is enforced independently of license — separation of concerns

8.3 API Key Management

- All third-party API keys should be stored as environment variables in production
- Keys are never logged or transmitted beyond their intended API endpoints
- Rate limiting handled gracefully — failed API calls return empty results without crashing

9. Future Development Roadmap

Phase 2 — Planned Features

Feature	Description	Priority
Cloud Processing	Offload heavy face recognition to cloud GPU instances	High
Batch URL Import	CSV/TXT import for bulk video download queues	High
Auto-Caption Generation	AI-generated captions via Whisper API	Medium
Video Thumbnail Generator	Auto-generate optimized thumbnails for each clip	Medium
Team Collaboration	Shared workspaces and job queues for teams	Medium
Mobile Companion App	iOS/Android app for remote monitoring of jobs	Low
Plugin Architecture	Allow third-party filter and export format plugins	Low
Analytics Dashboard	Advanced reporting with charts, trends, and export to PDF	Low

10. Conclusion

Content Creation Tool represents a significant advancement in desktop-based video content automation. By consolidating video acquisition, intelligent segmentation, AI-powered filtering, stock media sourcing, and secure user management into a single unified platform, the application substantially reduces the time and technical overhead required for professional video content production.

Phase 1 delivers a fully functional, production-ready application with a robust foundation for future enhancements. The modular architecture ensures that new features, integrations, and platform targets can be added with minimal disruption to the existing codebase.

The integration of multiple AI technologies — MediaPipe face detection, face_recognition deep learning, and Gemini AI ranking — positions Content Creation Tool as a genuinely intelligent content creation tool, not merely a workflow aggregator. This intelligent layer is the platform's primary competitive differentiator.

With the licensing and authentication infrastructure in place, Content Creation Tool is ready for commercial deployment and multi-user distribution.

Content Creation Tool — Built for Creators, Powered by AI
